

1.1 Structure & Calculation

AQA GCSE Mathematics 8300 — Number — spec refs N1–N5 — Papers 1 (non-calculator), 2 and 3 (calculator)

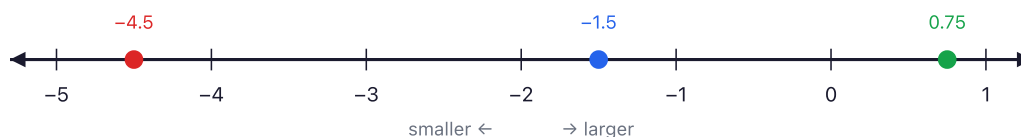
This topic is the foundation of every other part of the course: putting numbers in order, calculating accurately with and without a calculator, using the correct order of operations, and working with primes, factors and multiples. Most of these skills are tested on Paper 1 (non-calculator), so written methods matter.

1.1.1 Ordering Numbers and Inequality Symbols (N1)

- $=$ equal to $\neq$ not equal to
- $<$ less than $>$ greater than
- $\leq$ less than or equal to $\geq$ greater than or equal to

The "mouth" of $<$ or $>$ always opens towards the larger number: $3 < 8$ and $-2 > -5$.

On a number line, numbers **increase to the right**. A negative number with a larger digit is further left, so it is *smaller*: $-7 < -3$.



$-4.5 < -1.5 < 0.75$: the further right, the larger the number.

Ordering decimals and fractions

- **Decimals:** line up the decimal points and pad with zeros so each has the same number of decimal places, then compare digit by digit from the left. E.g. $0.5, 0.45, 0.405 \rightarrow 0.500, 0.450, 0.405$.
- **Fractions:** either convert to decimals, or write over a common denominator.
- **Mixed lists:** convert everything to decimals first.

Worked Example:

Write in order, smallest first: $\frac{3}{8}$, 0.38 , -0.4 , $-\frac{3}{7}$

$\frac{3}{8} = 0.375$ and $-\frac{3}{7} = -0.428\dots$

Negatives first (the "more negative" one is smaller): $-\frac{3}{7} < -0.4$. Then $0.375 < 0.38$.

Answer: $-\frac{3}{7}$, -0.4 , $\frac{3}{8}$, 0.38

Exam Tip: A very common error is thinking $0.45 > 0.5$ because "45 is bigger than 5". Always compare place value column by column: 4 tenths is less than 5 tenths.

1.1.2 The Four Operations and Place Value (N2)

Place value

Millions	Hundred thousands	Ten thousands	Thousands	Hundreds	Tens	Units	.	Tenths	Hundredths	Thou
3	0	7	4	5	1	2	.	6	8	9

In 3 074 512.689 the 7 is worth 70 000 and the 8 is worth $\frac{8}{100} = 0.08$.

Negative numbers

Multiplying and dividing: same signs $\Rightarrow$ positive; different signs $\Rightarrow$ negative.

$$(-6) \times (-7) = 42 \quad (-48) \div 6 = -8 \quad (-2)^3 = -8$$

Adding and subtracting: subtracting a negative is the same as adding: $-5 - (-8) = -5 + 8 = 3$.

Formal written methods

You must be able to use column addition and subtraction, long multiplication (or the grid method) and short or long division with integers, decimals, fractions and mixed numbers, all positive or negative.

Worked Example:

Work out 3.7×2.6 without a calculator.

Ignore the decimal points: $37 \times 26 = 740 + 222 = 962$.

There are $1 + 1 = 2$ decimal places in the question, so the answer has 2 decimal places: **9.62**. Check: $4 \times 3 = 12$, so about 10 is sensible.

Worked Example:

Work out $14.4 \div 0.6$.

Multiply both numbers by 10 so you divide by a whole number: $144 \div 6 = \mathbf{24}$.

Dividing by a decimal: multiply *both* numbers by the same power of 10 until the divisor is a whole number. The answer does not change:

$$7.2 \div 0.08 = 720 \div 8 = 90$$

Using a given fact

If you are given $23 \times 47 = 1081$, count how many times each number has been divided (or multiplied) by 10:

- 2.3×4.7 : both $\div 10$, so answer $\div 100 = 10.81$
- $1081 \div 4.7$: $1081 \div 47 = 23$, and dividing by a number 10 times smaller gives an answer 10 times bigger $= 230$

Household finance vocabulary

Term	Meaning
Cost price	The price a seller pays for an item
Selling price	The price the item is sold for
Profit / loss	Selling price – cost price (a negative result is a loss)
Debit / credit	Money taken out of / paid into an account
Balance	The amount in an account (negative if overdrawn)
Income tax	Tax on earnings above a tax-free personal allowance
VAT	Value Added Tax, added to the price of many goods (usually 20%)
Interest rate	The percentage paid on savings or charged on loans each year

Worked Example:

A trader buys 24 boxes of mangoes at £7.50 each. She sells 20 boxes at £11 and the rest at £6. Find her profit.

$$\text{Cost price} = 24 \times 7.50 = \text{£}180. \text{ Income} = 20 \times 11 + 4 \times 6 = 220 + 24 = \text{£}244.$$

$$\text{Profit} = 244 - 180 = \text{£}64.$$

Worked Example:

Priya earns £34 500 a year. The first £12 570 is tax-free; the rest is taxed at 20%. Find her monthly take-home pay.

$$\text{Taxable: } 34\,500 - 12\,570 = 21\,930. \text{ Tax: } 0.2 \times 21\,930 = \text{£}4386.$$

Take-home: $34\,500 - 4386 = 30\,114$ per year $\Rightarrow 30\,114 \div 12 = \text{£}2509.50$ per month.

1.1.3 Order of Operations and Inverse Operations (N3)

BIDMAS gives the priority of operations:

Brackets $\rightarrow$ **I**ndices (powers and roots) $\rightarrow$ **D**ivision and **M**ultiplication (left to right) $\rightarrow$ **A**ddition and **S**ubtraction (left to right)

Worked Example:

Work out $8 - 3 \times (-2)^2 + \sqrt{36}$.

Indices and roots: $(-2)^2 = 4$, $\sqrt{36} = 6$. Multiply: $3 \times 4 = 12$.

Then left to right: $8 - 12 + 6 = 2$.

Worked Example:

Work out $12 - 18 \div 3 \times 2$.

Division and multiplication left to right: $18 \div 3 = 6$, then $6 \times 2 = 12$. So $12 - 12 = 0$.

(Doing 3×2 first would give $12 - 3 = 9$, which is wrong.)

Inverse operations and reciprocals

- $+$ and $-$ are inverses; $\times$ and $\div$ are inverses; squaring and square rooting are inverses.
- The **reciprocal** of n is $\frac{1}{n}$. The reciprocal of $\frac{a}{b}$ is $\frac{b}{a}$. A number multiplied by its reciprocal equals 1.
- Dividing by a number is the same as multiplying by its reciprocal: $x \div 4 = x \times \frac{1}{4}$.
- Use cancellation to simplify, e.g. $\frac{36 \times 15}{9 \times 5} = 4 \times 3 = 12$.

Exam Tip: Use inverse operations to check answers on the non-calculator paper: if $1824 \div 16 = 114$, then 114×16 should give 1824. A fraction bar acts like brackets: work out the whole numerator and the whole denominator before dividing.

1.1.4 Primes, Factors and Multiples (N4)

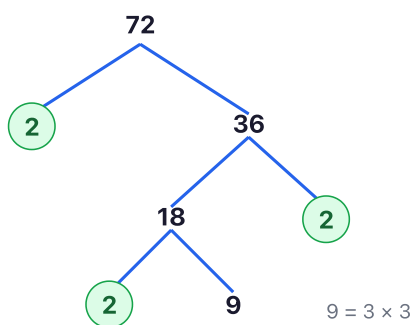
- Factor (divisor):** divides exactly into a number. Factors of 18: 1, 2, 3, 6, 9, 18.
- Multiple:** a number in the times table. Multiples of 6: 6, 12, 18, 24, ...

- **Prime number:** has exactly two factors, 1 and itself. 1 is *not* prime; 2 is the only even prime.
- **HCF:** highest common factor — the largest number that divides into all the numbers.
- **LCM:** lowest common multiple — the smallest number that all the numbers divide into.

Primes below 50: 2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47.

Prime factor decomposition

Every whole number greater than 1 can be written as a product of primes in exactly one way (the **unique factorisation theorem**). Use a factor tree or repeated division and write the answer in **index form**.



Factor tree for 72: circle the primes. $72 = 2 \times 2 \times 2 \times 3 \times 3 = 2^3 \times 3^2$.

HCF and LCM using prime factors

HCF = multiply the primes that appear in *both*, using the **lowest** power of each.

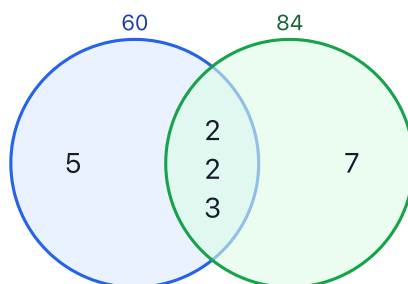
LCM = multiply *every* prime that appears, using the **highest** power of each.

Worked Example:

Find the HCF and LCM of 60 and 84.

$$60 = 2^2 \times 3 \times 5 \quad 84 = 2^2 \times 3 \times 7$$

$$\text{HCF} = 2^2 \times 3 = \mathbf{12} \quad \text{LCM} = 2^2 \times 3 \times 5 \times 7 = \mathbf{420}$$



Venn diagram of prime factors. HCF = product of the overlap ($2 \times 2 \times 3 = 12$); LCM = product of everything ($5 \times 2 \times 2 \times 3 \times 7 = 420$).

Worked Example:

Bus A leaves every 15 minutes and bus B every 25 minutes. Both leave at 9:00 am. When do they next leave together?

$15 = 3 \times 5$, $25 = 5^2$, so $\text{LCM} = 3 \times 5^2 = 75$ minutes.

9:00 am + 75 minutes = **10:15 am**.

Worked Example:

A floor 3.6 m by 2.4 m is covered with identical square tiles (whole number of cm, no cutting). Find the largest possible tile.

Work in cm: HCF of 360 and 240. $360 = 2^3 \times 3^2 \times 5$, $240 = 2^4 \times 3 \times 5$, $\text{HCF} = 2^3 \times 3 \times 5 = 120$.

Largest tile: **120 cm** square; it needs $3 \times 2 = 6$ tiles.

Exam Tip: Decide which you need from the context. **HCF:** splitting things into the largest equal groups or sizes. **LCM:** when events next happen at the same time, or the smallest amount that can be bought in two pack sizes. A number with exactly three factors is the square of a prime (e.g. 49 has factors 1, 7, 49).

1.1.5 Systematic Listing (N5)

To list all possible outcomes or combinations, work in a fixed order so nothing is missed or repeated. Lists, two-way tables and tree-style diagrams all help.

Worked Example:

List all the 3-digit numbers that use each of 2, 5 and 8 exactly once.

Fix the first digit, then swap the other two:

2 first: 258, 285 | 5 first: 528, 582 | 8 first: 825, 852

There are **6** numbers; 4 of them are greater than 500.

Shirt \ Shorts	Black (B)	White (W)
Red (R)	R, B	R, W
Blue (U)	U, B	U, W

Green (G)	G, B	G, W
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A table lists every combination of 3 shirts and 2 pairs of shorts: $3 \times 2 = 6$ outfits.

The product rule for counting Higher

If one choice can be made in m ways and a second, independent choice in n ways, then together there are

$$m \times n \text{ ways.}$$

This extends to any number of choices: $m \times n \times p \times \dots$

- **Repeats allowed:** the number of options stays the same each time, e.g. a 4-digit PIN: $10 \times 10 \times 10 \times 10 = 10\,000$.
- **No repeats:** the number of options drops by one each time, e.g. 4 different digits: $10 \times 9 \times 8 \times 7 = 5040$.
- **Arranging n different objects in a line:** $n \times (n - 1) \times \dots \times 2 \times 1$, e.g. 5 people: 120 ways.
- **Restrictions first:** deal with any restricted position before the others.

Worked Example:

A number plate is 2 letters (A–Z) then 3 digits; repeats are allowed but the first digit cannot be 0. How many plates are possible?

Letters: 26×26 . Digits: $9 \times 10 \times 10$ (only 9 choices for the first digit).

Total = $26 \times 26 \times 9 \times 10 \times 10 = \mathbf{608\,400}$.

Worked Example:

How many 4-digit PINs have all different digits and are even?

Restricted position first: the last digit must be 0, 2, 4, 6 or 8 $\rightarrow$ 5 ways.

The other three positions use the remaining 9 digits with no repeats: $9 \times 8 \times 7 = 504$.

Total = $5 \times 504 = \mathbf{2520}$.

Worked Example:

A lunch is either a main only (5 choices), a main and dessert (4 desserts), or starter, main and dessert (3 starters). How many lunches?

When the options are *alternatives* ("or"), add the separate counts:

$$5 + 5 \times 4 + 3 \times 5 \times 4 = 5 + 20 + 60 = 85.$$

Exam Tip: "And" means multiply (product rule); "or" between separate cases means add. Read carefully whether repeats are allowed — it is the most common source of lost marks in counting questions.

Summary

Skill	Key point
Ordering (N1)	Convert to decimals; negatives further left are smaller; use $<$, $>$, $\leq$, $\geq$, $\neq$
Four operations (N2)	Same signs positive, different signs negative; count decimal places; make the divisor whole
Order of operations (N3)	BIDMAS; $\div$ and $\times$ left to right; reciprocal of n is $\frac{1}{n}$
Primes and factors (N4)	Prime factors in index form; HCF = lowest powers of shared primes; LCM = highest powers of all primes
Listing (N5)	List systematically; product rule $m \times n$ Higher